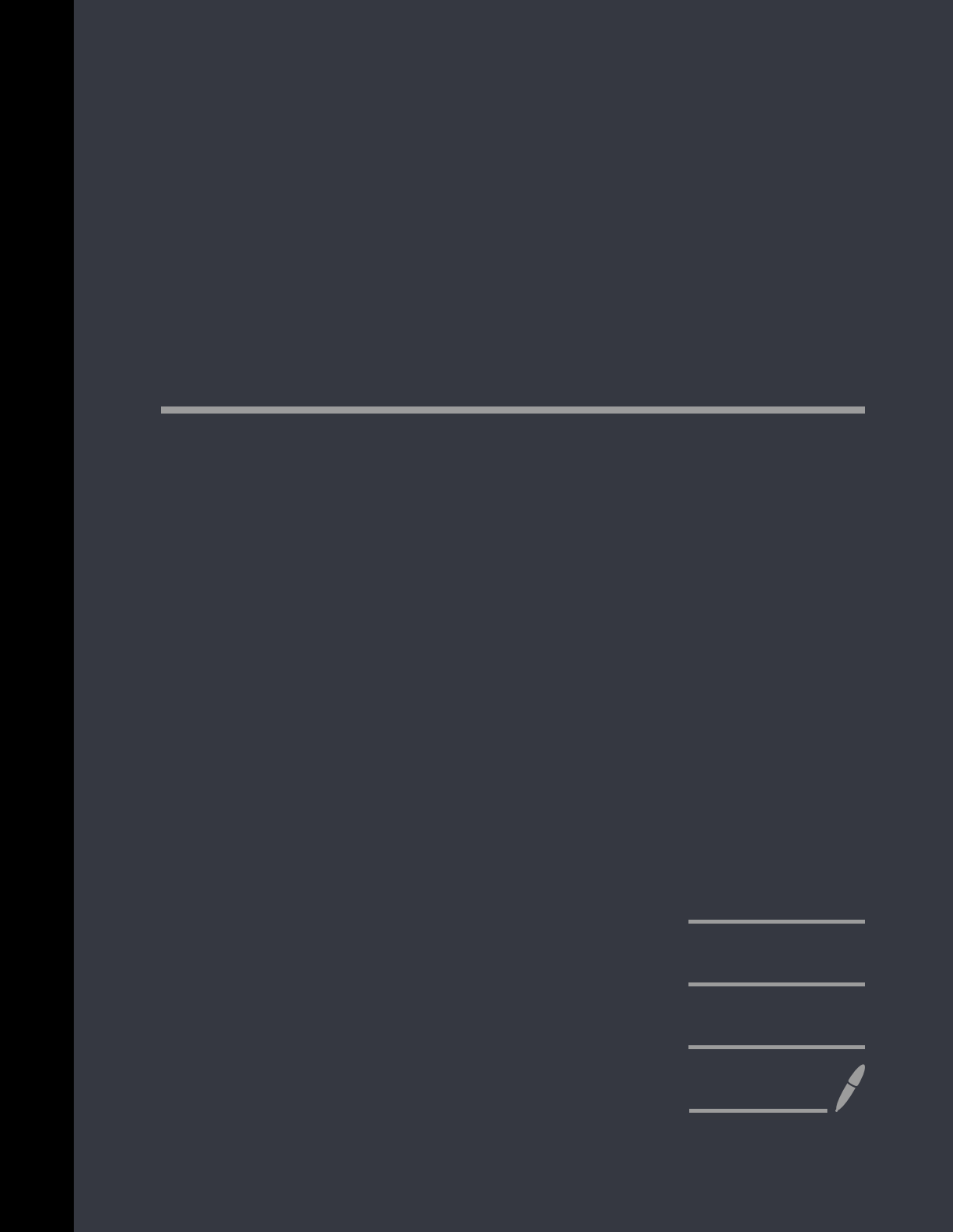


9) Es ist nicht Planar, da es $K_{3,3}$ als Minor ist

10)

A	B	C	$(A \to B) \leftrightarrow \bar{C} \to A)$
1	0	0	0
0	0	1	1
1	1	1	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$KNF = (A \vee B \vee C) \wedge$

$(A \vee \bar{B} \vee C) \wedge (\bar{A} \vee B \vee C) \wedge$

$(A \vee B \vee \bar{C}) \equiv$

$\equiv (A \vee C) \wedge (\bar{A} \vee B)$

$DNF = (\bar{A} \wedge \bar{B} \wedge C) \vee (\bar{A} \wedge B \wedge C) \vee$

$(A \wedge B \wedge \bar{C}) \vee (A \wedge B \wedge C) \equiv$

$\equiv (\bar{A} \wedge C) \vee (A \wedge B)$

11) $\overline{((\overline{C \setminus A}) \setminus B) \cap \bar{A}} = A \cup B \cup C$

$\overline{(\overline{C \cap \bar{A}}) \cap \bar{B} \cup A} = A \cup B \cup C$

$(C \cap \bar{A}) \cup B \cup A = A \cup B \cup C$

$(C \cap \bar{A}) \cup A \cup B = A \cup B \cup C$

$((C \cap A) \cap (A \cup \bar{A})) \cup B = A \cup B \cup C$

$C \cup A \cup B = A \cup B \cup C$

$A \cup B \cup C$

12) (R) $x \sim x \quad x^2 - x^2 = 0 = 2x - 2x$

(S) $x \sim y \Rightarrow x^2 - y = 2x - 2y \Rightarrow y^2 - x^2 = 2y - 2x \Rightarrow y \sim x$

(T) Sei $x \sim y$ und $y \sim z$ Dann gilt:

$x^2 - y^2 = 2x - 2y \qquad y^2 - z^2 = 2y - 2z$

Additionsverfahren: $x^2 - y^2 + y^2 - z^2 = 2x - 2y + 2y - 2z$

$x^2 - z^2 = 2x - 2z$

b)

$0^2 - 0^2 = 4 \cdot 0 - 2 \cdot 0 \qquad 1^2 - 1^2 = 2 \cdot 1 \cdot 2 \cdot 1$

$0 = 0 \qquad 0 = 0$

13) (I.A) $n=1$ (I.S) $n \to n+1$

$a_1 = 2$

$a_1 = \frac{1+1}{2} = 1$

$a_{n+1} = 2 - \frac{1}{a_n} = 2 - \frac{1}{\frac{n+1}{n}} = 2 - \frac{n}{n+1}$

$= \frac{2(n+1)}{n+1} - \frac{n}{n+1} = \frac{2n+2-n}{n+1} =$

$= \frac{n+2}{n+1}$

14)

I.A $n=2$

$\sum_{k=2}^{2} \frac{1}{(n+1)(k+1)} = \frac{1}{(2+1)(2+1)} = \frac{1}{3}$

$\frac{3}{4} - \frac{2 \cdot 2 + 1}{2 \cdot 2 \cdot (2+1)} = \frac{3}{4} - \frac{5}{12} = \frac{1}{2}$

(I.S) $n \to n+1$

$\sum_{k=2}^{n+1} \frac{1}{(k-1)(k+1)} = \sum_{k=2}^{n} \frac{1}{(k-1)(k+1)} + \frac{1}{n(n+2)} = \frac{2}{}$

15a) IA: $s=0$, $k=1$

$$(2-1)\cdot(2k-1)=(1-1)\cdot(2-1)=0\cdot 1=0 \quad \text{also } s$$

b) I.S:

$$s'=s+4k-1=(k-1)\cdot(2k-1)+4k-1=2k^2-k-2k+1+4k-1=$$

$$=2k+k=2(k'-1)+(k'-1)$$

$\underline{n=k+1}$ $\underline{s'(n)=2n^2+n}$